Hanyi Seok

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Research Interests: Carbon capture and conversion; electrochemical separations and resource recovery; reaction–transport modeling and measurement design

EDUCATION

Dual-degree program between Stony Brook University and the University of Seoul

Stony Brook University *Expected May 2027*
B.E. in Engineering Science, specialization in Environmental Engineering
Second Major in Applied Mathematics and Statistics
GPA: 3.97/4.00

University of Seoul *Expected May 2027*
B.E. in Environmental Engineering
GPA: 4.10/4.50

RESEARCH EXPERIENCE

Stony Brook University — Seo Research Group **Stony Brook, NY**
Undergraduate Researcher | Advisor: Prof. Hyowon Seo *May 2025 – Present*

Hollow-Fiber Membrane Contactor Design and Modeling [Project PDF]

- Designed and fabricated a hollow-fiber membrane contactor prototype for alkaline CO_2 absorption using Fusion 360 and polypropylene 3D printing; adjusted printing parameters and assessed sealing, tubing, and fiber-potting options.
- Developed a radial–axial finite-volume model using dissolved inorganic carbon and alkalinity as conserved transport variables; recovered speciation and pH from equilibrium while coupling gas-phase CO_2 depletion and tank balances. Checked carbon conservation, transport-operator convergence, and mesh and time-step sensitivity.
- In a separate lumped-model analysis, compared operating points with similar target-pH times and quantified their different responses to increased gas flow versus apparent mass-transfer capacity.

NTU-Based Analysis of CO_2 Loading Time [Project PDF]

- Derived an equilibrium-aware NTU expression for time to a target pH by coupling gas depletion, carbonate equilibrium, and the recirculating-tank balance.
- Compared the expression with perfect-sink predictions and a finite-pass numerical reference under matched transfer coefficients; identified increasing contributions from within-pass liquid loading at lower liquid flows and near-equilibrium targets.
- Derived a first-order material-balance correction for within-pass liquid loading; relative to the uncorrected expression, it reduced absolute time discrepancies from the finite-pass numerical reference in all 100 tested operating-condition and target combinations.

Northeastern University — Barecka Lab [Project PDF] **Remote**
Undergraduate Researcher | Advisor: Prof. Magda H. Barecka *Jun 2026 – Aug 2026*

- Implemented compartmental carbon and isotope balances for anion-exchange-membrane gas-diffusion-electrode reactors to model ^{13}C enrichment in unreacted CO_2. Included product-carbon inventories and interfacial isotope exchange at zero net total-carbon flux.
- Fitted a reduced absorption model to 27 archived zero-current bubbling conditions (median within-sample pH RMSE: 0.180), assessing residuals and parameter bounds; separately evaluated baseline sensitivity in archived PTR-MS-derived outlet-isotope responses.
- Analyzed published anode-gas data using a reduced bicarbonate–carbonate relation; assumptions of 100% and 95% oxygen Faradaic efficiency produced similar fits but different effective current-sharing factors. Proposed independent oxygen, charge, and liquid-carbon measurements to test the interpretation.
- Implemented and evaluated eight effective-parameter estimation procedures in Python/Jupyter using synthetic data, with parameter-recovery tests, held-out predictions, and local rank and uncertainty checks; identified weak practical separation of membrane current-carrier weights.

Korea Institute of Energy Technology — Wooyul Kim Lab [Project PDF] **Naju, South Korea**
Undergraduate Researcher | Advisor: Prof. Wooyul Kim *Jul 2026 – Aug 2026*

- Prepared metal-modified CeO_2 nanorod supports by alkaline hydrothermal synthesis and deposited nominally 2 wt% Ru via $NaBH_4$ reduction for a team study of photothermal CO_2 hydrogenation.
- Screened seven CeO_2 support modifiers using photothermal reaction tests and GC; Ru/Zr–CeO_2 gave the highest CH_4 response in the initial batch, with approximately 20% higher peak area than Ru/CeO_2 after 1 h.

- Used GC results to select follow-up Zr compositions up to a nominal 30 mol%; compared responses within each batch, with the highest-response composition at the upper tested limit.

Rutgers University — Dismukes Research Group [Project PDF] **Remote**
Undergraduate Researcher | Advisor: Prof. G. Charles Dismukes *Jun 2026 – Aug 2026*

- Developed a reproducible Python workflow for 116 microscopy images from 20 Bi-deposited Toray electrodes and five pristine Toray specimens across four magnifications; aggregated repeated views by specimen and magnification to avoid pseudoreplication.
- Applied specimen-level permutation tests, bootstrap confidence intervals, and Holm correction, identifying higher mean grayscale intensity at 260× (adjusted $p = 0.0024$) and a smaller P95–median intensity gap across all four magnifications.
- Built an exploratory candidate-detection pipeline for 260× carbon-fiber images using local z-score seeding and marker-controlled watershed; generated candidate counts, pixel-scale morphology summaries, and quality-control flags.

University of Seoul — Water Environment and Engineering Lab [Project PDF] **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Ihnsup Han *May 2024 – Jan 2025*

- Designed and operated a laboratory-scale batch reactor combining electrochemical treatment and granular activated-carbon adsorption for synthetic monoethanolamine (MEA) wastewater relevant to CO_2 capture.
- Compared electrode materials (stainless steel, Ti, Al), voltage, spacing, carbon dosage, initial MEA concentration, and temperature; monitored MEA removal and nitrogen species to examine potential removal pathways.
- Combined initial screening, follow-up ANOVA/Tukey comparisons, literature evidence, and electrode costs to recommend operating conditions within the tested range, including Al electrodes for subsequent trials.

University of Seoul — Environmental Systems Toxicology Lab [Project PDF] **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Jinhee Choi *Jun 2021 – Feb 2022*

- Integrated ToxCast and ToxRef records spanning 449 chemicals and 949 assays by CAS number; distinguished inactive assay results from missing observations and aligned in vitro activity with rat developmental and reproductive toxicity records.
- Performed endpoint-specific point-biserial correlations in R using binary activity and $\log_{10}$-transformed ToxRef values; shortlisted eight associations ($|r| \geq 0.5$; $n \geq 15$) and selected two GPCR-target assays for biological review.

SELECTED PROJECTS

Dissolved Oxygen Prediction for Anyang Stream [Project PDF]
Smart Water Environment Management *Sep 2024 – Dec 2024*

- Developed a dissolved oxygen prediction workflow using 17,965 water-quality records and five predictors; selected MICE–KNN ensemble imputation after comparing four imputation methods under 10–30% simulated missingness.
- Compared six predictive models and tuned an LSTM network, achieving $R^2 = 0.915$ on the test set and 0.966 on subsequent November 2024 observations not used for training.
- Ranked first among all course projects and presented the project at the course showcase.

Domestic SAF Roadmap for Korea [Project PDF]
Unit Design of Processes in Environmental Engineering *May 2024 – Jul 2024*

- Led an eight-member team in comparing HEFA, ATJ, FT, and PtL routes for a Korean SAF supply roadmap.
- Estimated annual potential from domestic feedstocks: 734.6 million L of bio-SAF using heating values and conversion efficiencies, and 268.0 million L of synthetic SAF using carbon-content calculations.
- Projected a synthetic-SAF shortfall of 254.3 million L in 2050 using 2009–2021 consumption trends and European blending benchmarks; examined DAC, hydrogen supply, and fuel-price scenarios. The project ranked first among eight teams.

University of Seoul Campus GHG Inventory and 2040 Net-Zero Roadmap [Project PDF]
Unit Design of Processes in Environmental Engineering *Mar 2024 – May 2024*

- Led an eight-member team in constructing a 2023 campus GHG inventory using institutional records and Korean/IPCC emission factors; estimated net emissions of approximately 14,900 tCO_2e after forest uptake.
- Analyzed 46 commuting-survey responses using mode-specific emission factors, round-trip distances, and annual commuting frequency; scaled group-level estimates to student, faculty, and staff populations.
- Proposed a phased roadmap linking renewable generation and microgrids with waste-reduction incentives and financing options, targeting a 50% reduction by 2030 and net zero by 2040; the project ranked first among eight teams.

PRESENTATIONS

H. Seok and D. Jeong. *Process Optimization for Treatment of Amine-Rich Wastewater Generated from Wet Carbon Capture.* Undergraduate Thesis Presentation, Urban Sciences College, University of Seoul, South Korea, 2025.

H. Seok, S. Jang, and Y. Baek. SELFIES ALPHA: An Index Based Notation Model with Optical Isomer Symbols for Improved Recognition Capacity within Toxicity Prediction. Poster, Environmental Health Policy Idea Competition, South Korea, 2023.

H. Seok, M. Jo, and S. Ahn. *Identification of ToxCast Assays for Reproductive Toxicity Screening by Correlation Analysis of ToxCast and ToxRef Data.* Research Workshop, Graduate School of Chemical Safety Management, University of Seoul, South Korea, 2022.

HONORS & AWARDS

Tau Beta Pi Engineering Honor Society *Fall 2025*

Dean's List, Stony Brook University *Spring 2025; Fall 2025; Spring 2026*

Korea–U.S. Advanced Technology Youth Exchange Program Scholarship (KIAT) *Spring 2025 – Fall 2025*

Presidential Scholarship, University of Seoul *Spring 2021 – Fall 2024*

Outstanding Student Scholarship, Digital Water Industry Innovation Talent Development Program *Spring 2024*

Gold Prize, 2023 Environmental Health Policy Idea Competition (1st of 15 teams) *Nov 2023*

Graduate School Dean's Award, University of Seoul (2nd of 20 teams) *Feb 2022*

TECHNICAL SKILLS

Programming: Python, R

Modeling & Simulation: Reaction–transport and compartmental modeling; finite-volume methods; parameter estimation and identifiability; sensitivity and uncertainty analysis

Data Analysis & Statistics: Statistical inference, LSTM, missing-data imputation, microscopy image analysis

Experimental Methods: Electrochemical wastewater treatment, catalyst synthesis and Ru deposition, photothermal reaction testing, gas chromatography, cyclic voltammetry

CAD & Fabrication: Fusion 360, SolidWorks, 3D printing

ADDITIONAL EXPERIENCE

Republic of Korea Army **Yangpyeong, South Korea**

Logistics Clerk | Squad Leader *Apr 2022 – Oct 2023*

- Developed Python scripts to clean, cross-check, and summarize administrative records for weekly reporting; automated printing of a logistics list containing more than 1,000 items using VBS.
- Led a 12-soldier squad, coordinating training schedules, safety procedures, personnel documentation, and performance evaluations.